

Pesticide toxicity and their impact on plant growth, active constituents and productivity

Hiyani Gautam^a, Shubhra Singh^a, Hema Prashad^a, Antul Kumar^b, Anuj Choudhary^b, Harmanjot Kaur^b, Om Prakash Narayan^c, Shambhu Krishan Lal^{d,e}, and Sahil Mehta^f

^aDepartment of Entomology, Dr. YS Parmar University of Horticulture and Forestry, Nauni Solan, Himachal Pradesh, India, ^bDepartment of Botany, Punjab Agricultural University, Ludhiana, Punjab, India, ^cBME Department, Tufts University, Medford, MA, United States,

^dSchool of Genetic Engineering, ICAR–Indian Institute of Agricultural Biotechnology, Ranchi, Jharkhand, India, ^eInternational Centre for Genetic Engineering and Biotechnology, New Delhi, India, ^fDepartment of Botany, Hansraj College, University of Delhi, New Delhi, India

Abbreviations

<i>A. cepa</i>	<i>Allium cepa</i>
ABA-3	abamectin
APOX	ascorbate peroxidase
B	boron
Ca	calcium
CAT	catalase
Co	cobalt
CO ₂	carbon dioxide
Cu	copper
ETL	economic threshold level
Fe	iron
GBH	glyphosate-based herbicides
GPOX	glutathione peroxidase
GR	glutathione reductase
GSH	reduced glutathione
GST	glutathione S-transferase
H ₂ O	water

H ₂ O ₂	hydrogen peroxide
IAA	indole-3 acetic acid
IPM	integrated pest management
K	potassium
<i>L. esculentum</i>	<i>Lycopersicon esculentum</i>
MDA	malondialdehyde
Mg	magnesium
Mn	manganese
Mo	molybdenum
O ₂	oxygen
P	phosphorus
<i>P. vera</i>	<i>Pistacia vera</i>
PGPRs	plant growth-promoting bacteria
POD	peroxidase
REI	restricted entry interval
RL	root length
ROS	reactive oxygen species
RSL	root shoot length
SL	shoot length
SOD	superoxide dismutase
THM-1	trihalomethanes
<i>V. radiate</i>	<i>Vigna radiate</i>
<i>Z. mays</i>	<i>Zea mays</i>
Zn	zinc

1. Introduction

Day by day growing population is exerting pressure on land and agriculture which is threatening food security (Mehta et al., 2019; Rahman et al., 2019; Singh et al., 2019; Kumar et al., 2021a, 2021c). Due to which the demand for food is continuously increasing which has forced human beings to utilize synthetic chemical pesticides for improving the crop (Mehta et al. (2020); Kumar et al., 2021b, 2021d; Tomer, 2013; Fig. 1). Pesticide is a hazardous chemical substance or biological compound that are significantly liberated into the surrounding to deter, avert and overall check the populations of weeds, insects, fungi, rodents, or other crop enemies (Homayoonzadeh et al., 2021). They seduce, attract and then destroy or eliminate the pests (Kaur et al., 2019).

Pest can be defined as “the plants or animals that jeopardize our food, health and/or comfort” (Kaur et al., 2019). Crop pathogens infect crops from field to post-harvest/storage. However, the pesticide is being considered as a safeguard of crops that protect the plant from qualitative and quantitative damage (Goh et al., 2011). Moreover, the usage of pest-resistant transgenic varieties and biopesticides helps to mitigate the challenges of crop pests. Though, chemical pesticides are the main line of defense in protecting the crops from the damage of insects or pests and the use of chemical pesticides also results in increase yield of crops (Kaur et al., 2019).

Historically, in 1962, a book named “Silent Spring” was published by Rachel Carson and in her book, she wrote about the detrimental effects of pesticides on the environment. After that, a lot of research was conducted in this regard (Casida and Quistad, 1998). The use of pesticides in agroecosystems has raised public concern due to the harmful impacts of these substances on human as well as environmental health. Furthermore, the phytotoxic effects

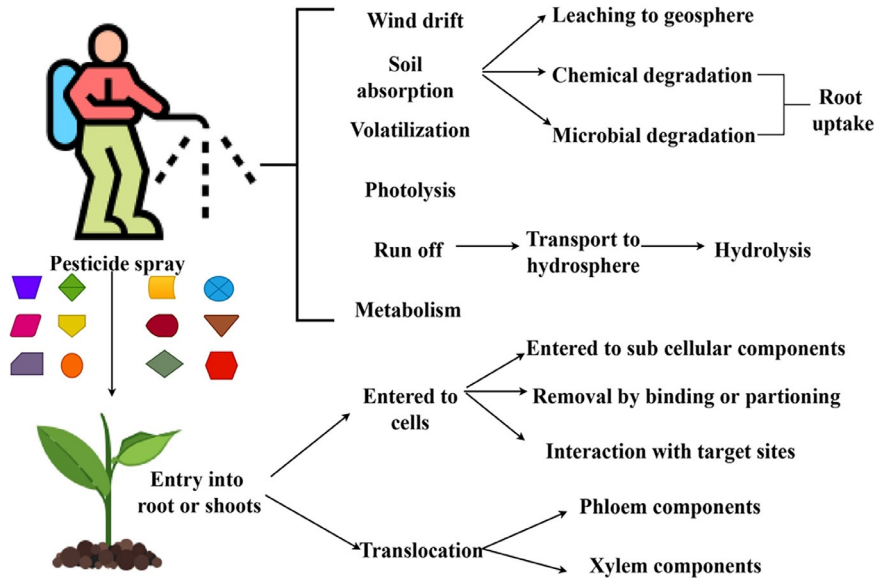


FIG. 1 Diagrammatic flowchart of flow and uptake of pesticides when applied in fields.

of some pesticides have also been recognized. As a source of carbon and nutrients, many beneficial soil microbes consume pesticides, furthermore decline soil health (Hassaan and El Nembr, 2020). Pesticides applications cause a deleterious effect on germination and photosynthetic pigments on various plants species such as using 100 mg/kg Chlorpyrifos to affect seed germination in *Cenchrus setigerus* (Dubey and Fulekar, 2011), Emamectin benzoate, Cypermethrin, Lambda-cyhalothrin, and Imidacloprid shows inhibitory effect in *Lycopersicon esculentum* (Shakir et al., 2016), Mancozeb and Chlorpyrifos ranges from 10 to 150 ppm negatively affect germination phase (Fatma et al., 2017). Pesticides like mancozeb and chlorpyrifos show the phytotoxic effect on growth and development of onion (Fatma et al., 2018) (Table 1).

2. Phytotoxicity effects on plants

The excessive application of pesticides causes contamination in plants that negatively regulating their cellular components. It significantly alters the respiration rate, leaf function, osmotic imbalance, hormonal signaling, and removal of wax and cuticle layer in the root system (Mehta et al., 2021, 2022). However, miscalculation and excessive doses of pesticides products used in reserve tanks can also enhance penetration rate into plant cells. (Snell Scientifics LLC, 2017). Besides these, environmental factors like temperature, humidity, and light play a crucial role in the likelihood of plant phytotoxicity whereas temperatures above 80°F alleviate phytotoxic injuries in plants. Several phytotoxicity symptoms such as chlorosis, leaf margin necrosis, leaf speckling, brown or leaf cupping ultimately may lead to plant death. For example, *Geranium* leaves exhibit phytotoxic symptoms from a chlormequat chloride-containing plant growth regulator. In another study, Aluthmuhandiram et al., 2021 demonstrated the

TABLE 1 Impacts of pesticides on growth and germination of different plants.

S. no.	Plant species	Pesticide used	Type of pesticide	Amount used	Mode of application	Effect of pesticide	References
Impact on germination							
1.	<i>Cucumis sativus</i> and <i>Zea mays</i>	Paraquat and Glufosinate-ammonium	Herbicides	Paraquat, glufosinate-ammonium at 200, 400, 600, and 800 g a.i. ha ⁻¹	Field	The germination rate of more than 90% was observed than untreated	Wibawa et al. (2009)
2.	<i>Cenchrus setigerus</i> , <i>Pennisetum pedicellatum</i>	Chlorpyrifos, Fenvalerate and Cypermethrin	Insecticides	0–100 mg/kg	Soil	Significant reduction and delay in seed germination	Dubey and Fulekar (2011)
3.	<i>Z. mays</i>	Pendimethalin	Herbicides	0–10 ppm	Hoagland's nutrient solution	Percent germination, length of plumule, and radicle decreased	Rajashekhar et al. (2012)
4.	<i>Lycopersicon esculentum</i>	Emamectin benzoate Cypermethrin Lambda-cyhalothrin Imidacloprid	Insecticides	Emamectin benzoate at 10, 20, 40, 80 and 160 mg/L Cypermethrin at 30, 60, 125, 250 and 500 mg/L Lambda-cyhalothrin at 15, 30, 56, 120 and 240 mg/L Imidacloprid at 125, 250, 500, 1000 and 2000 mg/L	Soaking of seeds in pesticide solutions	Up to 3 days after pesticide application the highest dosages of all the pesticides almost completely inhibited the germination and in later days also germination was lower when compared with control.	Shakir et al. (2016)
5.	<i>Brassica juncea</i>	Mancozeb	Fungicide	1, 2 and 4 mg/kg	Seed treatment	The decline of 30%–60% and 17%–55% was recorded at the application rate of 2 and 4 mg/kg in RH-30 and Laxmi variety, respectively, in comparison with control	Monika and Kidwai (2017)

6.	<i>Allium cepa</i>	Mancozeb and chlorpyrifos	Fungicide and insecticide	Mancozeb at 10, 30, 50, 70, 90, 110, 130 and 150 ppm Chlorpyrifos at 5, 10, 15, 20, 25 and 30 ppm	Seed treatment	Seed treatment with 150 ppm Mancozeb decreased germination, up to 41.66%; germination of 30 ppm chlorpyrifos treated seeds show an 18.33% decline	Fatma et al. (2017)
Impact on chlorophyll content							
7.	<i>L. esculentum</i>	Emamectin benzoate Cypermethrin Lambda-cyhalothrin Imidacloprid	Insecticides	Emamectin benzoate at 10, 20, 40, 80 and 160 mg/L Cypermethrin at 30, 60, 125, 250 and 500 mg/L Lambda-cyhalothrin at 15, 30, 56, 120 and 240 Imidacloprid at 125, 250, 500, 1000 and 2000	Soaking of seeds in pesticide solutions	With the increase in pesticide dose, total chlorophyll content got decreased	Shakir et al. (2016)
8.	<i>Vigna radiata</i> L.	Chlorpyrifos	Insecticides	0.6 and 1.5 mM	Foliar application	20% decline in photosynthetic pigments at 1.5 mM chlorpyrifos	Parween et al. (2011a,b)
9.	<i>L. esculentum</i>	Acetamiprid, Imidacloprid, Abamectin, Thiamethoxam, Abamectin+ Chlorantraniliprole	Insecticides	Acetamiprid @ 30 mg, Imidacloprid @ 100 mg Abamectin @ 25 mg Thiamethoxam @ 100 mg and Abamectin + Chlorantraniliprole @ 90 mg in 100 L water	Spray	Decreased total chlorophyll content and net photosynthetic rate	Yildiztekin et al. (2019)
10.	<i>B. juncea</i>	Mancozeb	Fungicide	1, 2 and 4 mg/kg	Seed treatment	Chlorophyll content reduced with the increase in dosages of Mancozeb	Monika and Kidwai (2017)
11.	<i>Pistacia vera</i> L.	Imidacloprid and phosalone	Insecticides	0.21 g a.i./l and 1.21 g a.i./l,	Spray	Little change was observed in total chlorophyll content	Homayoonzadeh et al. (2021)

temperature effects on phytotoxin-induced virulence in *Lasiodiplodia theobromae*. Findings suggested that an increase in temperature causes significant alleviates phytotoxicity of detached leaves of the grapevine. Moreover, temperature above 35°C also causes major necrotic leaf lesions, if inoculated with crude culture filtrates (Aluthmuhandiram et al., 2021).

Recent studies demonstrated that leaf miner infestations in Poke Choy and Correnta in spinach could be detected accurately by using the Hyperspectral remote sensing technique (Nguyen and Nansen, 2020). Visual phytotoxicity inspection of three ornamental plants, i.e., *Zinnia elegans*, *Tagetes patula*, and *Gerbera* sp. was done for detection of a numbered phytotoxic compound (V-10433) which was applied at different doses (Nansen et al., 2021).

3. Pesticides phytotoxicity symptoms on plants

Plants having insecticide phytotoxicity include the following symptoms: Abnormal Growth, Chlorosis, Leaf Distortion, Stunting, Necrosis, Vein clearing, Epinasty, Hyponasty, etc. (Shakir et al., 2016; Homayoonzadeh et al., 2020).

For example, studies conducted by Ebel et al. (2000) observed the imidacloprid phytotoxicity on cucumber and tomato in the controlled conditions. In tomato (*L. esculentum* Mill. "Rutgers") and cucumber (*Cucumis sativus* L. "Turbo"), exogenous application of Marathon (imidacloprid) using of at 0, 1/8, 1/4, 1/2, 3/4, and 1 tsp. (0, 5, 10, 20, 30, and 40 mg a.i.) per 4.5 in. (550 mL) pot. Results of their study revealed that phytotoxic symptoms appear on various imidacloprid-treated plants within 6–7 days after application. On both tomato and cucumber plants phytotoxicity, signs of leaf chlorosis appeared primarily on the oldest leaves, whereas young leaves show marginal necrosis and distorted growth within 1 week after application. The above findings suggested that the lowest application rate has resulted in the development of phytotoxicity symptoms in both plants (Fatma et al., 2017; Yildiztekin et al., 2019; Homayoonzadeh et al., 2021). Visual symptoms were somewhat similar to Ca (calcium) deficiency symptoms but after the foliar analysis of cucumber no difference was found in Ca (Calcium), Co (Cobalt), Fe (Iron), Zn (Zinc) amid rates of imidacloprid insecticide. Although the concentration of B (Boron) and Mg (Magnesium) reduced whereas, Mn (Manganese) and K (Potassium) linearly enhanced with imidacloprid rates.

Cu (Copper), Mo (Molybdenum), and P (Phosphorus) altered quadratically using 20 mg a.i. or 1/2 tsp. per pot having the reduced level of Cu, Mo, and P and that observed to be alleviating at the greater rates. Results concluded that imidacloprid can adjust nutrition in the plant. On the contrary, Hasan et al. (2016) used the tomato plants to evaluate the indoxacarb effect against *Helicoverpa armigera* Hub. (tomato fruit borer) and induce phytotoxicity (Monika and Kidwai, 2017; Fatma et al., 2017; Yildiztekin et al., 2019). Indoxacarb was applied at application rates of 30, 40, 50, 60, and 75 g a.i./ha on tomato plants. However, treatment with indoxacarb such as 150 and 250 g a.i./ha concentration doesn't show any phytotoxic effect.

Similarly, another study conducted by Balikai and Mallapur (2017) revealed that flubendiamide applied at different concentrations did not show any visible phytotoxic symptoms on gherkin plants at 1, 3, 5, 7, 10, and 15 days. According to Homayoonzadeh et al. (2021), insecticides like Phosalone and imidacloprid of concentration 0.21 g a.i./l and 1.21 g a.i./l, respectively, affect chlorophyll content in *Pistacia vera*. In another study, foliar application

of various pesticides such as Acetamiprid, Imidacloprid, Abamectin, Thiamethoxam, Abamectin+Chlorantraniliprole causes a reduction in chlorophyll content in *L. esculentum* (Yildiztekin et al., 2019). Despite these, several reports also mentioned that the application of pesticides has adverse effects on plant physiology, including alteration of nitrogen content, carbon metabolism, reduction in growth, perturbation flower development, and/or (Parween et al., 2011a,b; Shakir et al., 2016; Homayoonzadeh et al., 2021).

4. Impact of pesticide toxicity on soil health and plant

4.1 Soil health

The pesticides are used to inhibit pest attack and increase productivity but pesticides show hazardous effects on the environment too. Due to their excessive pesticide usage and its residues significantly accumulate in the plant and soil causes an enhanced contamination rate, which affects biodiversity directly or indirectly (Biondi et al., 2012; Han et al., 2017). Pesticides can adversely affect the biological and physico-chemical properties of the soil (Reddy et al., 2021). Moreover, pesticides can also disrupt the activities of soil microbes (Nannipieri et al., 2013).

Al-Ani et al. (2019) observed the impact of two insecticides, Malathion 50% and Alphacypermethrin 10% WP on soil microbes including bacteria, fungi other actinomycetes, and production of CO₂ (carbon dioxide). Results confirmed that production of CO₂ was significantly reduced for malathion and Alphacypermethrin. However, observation at 42–45 days shows that values for the production of CO₂ were 36% and 32% using Alphacypermethrin concentrations 200 and 100 ppm, respectively. Besides these, the application of Malathion (50, 100, and 200 ppm) lowered CO₂ production by 42%, 45%, and 52%, respectively. Additionally, microorganisms' concentration and their activity depend upon the insecticide's dosage applied to the soil samples. Another, Baćmaga et al. (2016) revealed the impact of fungicides like Falcon 460 EC in soil that significantly suppresses the various enzymatic activity of urease, catalase, dehydrogenase, and alkaline phosphatase. Furthermore, upregulation of active ingredients causes deleterious effects on soil residing in microbes communities. Similarly, Baćmaga et al. (2018) also stated the influence of Chlorothalonil on the biochemical properties and soil-borne microorganisms.

4.2 Plant's growth and active constituents

Seed's treatment using pesticides positively helps in protecting seedlings from the pathogen infestation via prevent the entry of pathogenic components and but negatively affect germination rate and seedling growth (Dhanamanjuri et al., 2013; Lal et al., 2018). Through transpiration, pesticides absorb from the soil into various parts of the plant (Sharma et al., 2019a,b; Lushchak et al., 2018). These ill effects of high dosage or nonselective pesticides can be visible and measured by the rate of burns, leaves twisting, chlorosis, necrosis, and stunting percentage (Sharma et al., 2015a,b,c; Sharma et al., 2018a). For instance, Parween et al. (2011a,b) using Chlorpyrifos (an organophosphate insecticide) for managing insects reported *Vigna radiata* L. that cause a reduction in nitrogen metabolism and plant growth. Additionally, Homayoonzadeh et al. (2020) sprayed dichlorvos and imidacloprid on *Cucumis*

sativus seedlings to control the damage of whiteflies. Results revealed that spraying both (dichlorvos and imidacloprid) pesticides directly enhanced the enzymatic activities including ascorbate peroxidase, catalase, guaiacol peroxide, phenylalanine ammonia-lyase, and superoxide dismutase in *C. sativus* seedlings. Furthermore, carbohydrates, proline, total protein, total phenolic content, and total soluble sugar content are also alleviated in a significant way. Interestingly, dichlorvos and imidacloprid treatments led to a reduction in the activity of polyphenol oxidase but show very few changes in malondialdehyde content, electrolyte leakage, and hydrogen peroxide in *P. vera* L. (Homayoonzadeh et al., 2021).

The use of pesticides has adverse effects on seed germination. Rajashekhar et al. (2012) conducted a study using pendimethalin on *Z. mays* L. and they observed maximum germination percentage (95%) under controlled conditions, whereas the drastic decline in germination percentage was measured with increasing dosage of pendimethalin. The reduction was recorded up to 69% at an increased amount of pendimethalin (> 10.0 ppm) (Table 1). Interestingly, which may be due to the injurious effect of pendimethalin causes adverse effects on seed reserves inhibits mobilization and degradation rate. Studies were conducted by Moore and Kroger (2010) to evaluate the impact of two herbicides and three insecticides on seedling of *Oryza sativa* germination based on exposure up to 4–5 days. In tested insecticides, diazinon had an improved germination percentage (85%) compared with control whereas fipronil has reduced seed germination (76%) when compared with control (80%). In the case of herbicide atrazine/metolachlor mixture, a better germination rate (81%) was recorded in comparison with single herbicide atrazine in the *O. sativa* seedlings. Inhibitory effects of pendimethalin and trifluralin were observed by (Nehru et al., 1999) on germination traits and initial growth in *Z. mays* and *V. radiata* L, respectively.

Similarly, Fatma et al. (2017) reported a remarkable reducing effect of pesticides on germination parameters, and their effect gets increased with an increase in pesticide dose (Table 1). Koornneef et al. (2002) revealed that seed germination is highly affected by environmental stress, Goswami et al. (2013) also reported that seed germination is very sensitive to pesticides. Root length, root/shoot length ratio, shoot length, and survival percentage of treated *Allium* seeds with various pesticides concentrations such as Mancozeb and Chlorpyrifos were observed lower than that of control exhibiting a dose-based reduction (Fatma et al., 2017). This proved that the nature of these pesticides (Mancozeb and Chlorpyrifos) is phytotoxic. Earlier (Nielson and Rank, 1994; Amin, 2002) also stated that growth traits like root length (RL), Root shoot length (RSL) and shoot length (SL) could be used to measure toxicity like pesticides. Reduction in plants growth was observed when pesticides were applied at higher concentrations which ultimately resulted in cell elongation, cell division reduction, and indole-3 acetic acid (IAA) conversion into different photo-oxidative products because these components act as IAA antagonists (Tevini and Teramura, 2008). In *Allium cepa* seedlings, significant variation in Seedling vigor index (SVI) was observed between different treatments for Mancozeb and Chlorpyrifos (Fatma et al., 2017). Similarly, Habtamu et al. (2013) also experimented on seedling growth and germination rate of haricot bean cultivars. The finding revealed that a significant dose-dependent decrease in shoot-root length was observed at higher levels of copper sulfate solution. Phytotoxic effects enhanced significantly which is an indicator of pesticidal toxicity used in their experiment.

Plants use their antioxidant enzyme system including SOD, CAT, and POD (superoxide, dismutase and catalase, peroxidase) (Jiang and Yang, 2009) (Fig. 2). Fatma et al. (2017) explained

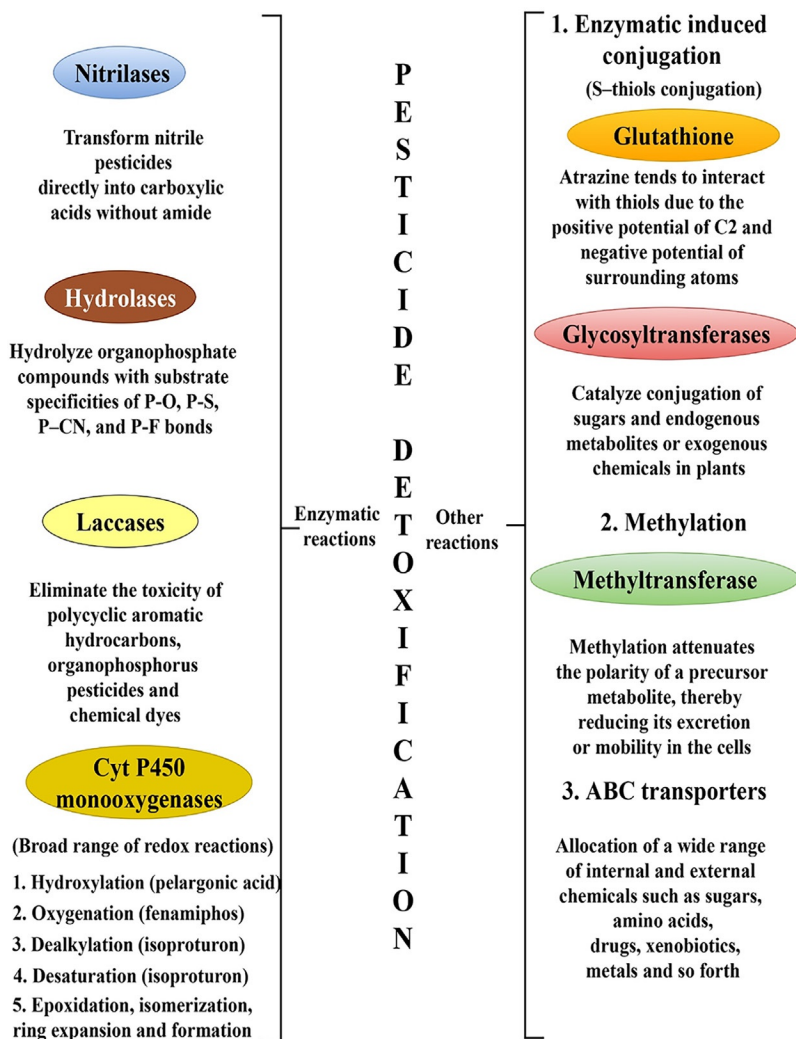


FIG. 2 Schematic presentation of pesticide detoxification by plants enzymatic and nonenzymatic machinery.

that the application of pesticides using *A. cepa* seedlings viz., Mancozeb, and Chlorpyrifos imposed oxidative stress on plants to environmental stress. Similar results also suggest the interaction between pesticides and the activity of the antioxidative enzymes. (Pandey and Rizvi, 2010; Jan et al., 2012) Oxidoreductase activity of CAT help in the liberation of H_2O_2 (hydrogen peroxide) by converting H_2O_2 to H_2O (water) and O_2 (oxygen). Zhang et al. (2009) demonstrated the same results in rice and wheat plants after application of 1,2,4-trichlorobenzene dimethoate (Mishra et al., 2009). In *Allium* seedlings, significant enhancement in CAT activity and increased in dose manner (Mei et al., 2009; Fatma et al., 2017).

Enhanced activities of SOD-like enzyme were observed in *Glycine max* L. under high insecticide dosage (Bashir et al., 2007). In wheat, herbicidal exposure indicates upregulation of

SOD activity was regulated by O^- , scavenging and causes damage in *Allium* seedling induced toxicity from chlorpyrifos and mancozeb (Jianga et al., 2010; Wu et al., 2010).

4.3 Plant production

There are many evident visual symptoms like necrosis, discoloration, and deformation, etc. and in many cases, effects on yield (weight and appearance) are also recorded upon used of pesticides. Some important effects may only become apparent after repeated treatments and not necessarily translate into visible necrosis (Ferree, 1979). In many cases, the phytotoxic effects of insecticide application result in the yield loss of the treated crop. Shahid et al., 2018 studies the glyphosate-based herbicides (GBH) effect on *Medicago sativa* L. The finding showed an alleviate in lipid peroxidation that causes the suppression of shoots and roots growth (Fernandes et al., 2021). Parveen (2012) conducted a study on mungbean seedling using a foliar application of chlorpyrifos. The results concluded that 0.3 mM chlorpyrifos exposure improved the yield associating traits, i.e., number of seeds per plant, the highest number of pods, and dry seed weight. However, higher leaf area, enhanced plant height, and more uptakes of nutrients would have increased the translocation of photosynthates which ultimately resulted in higher pods number per plant.

Application of Chitosan at the early phase of growth promoted seed formation in soybean and rice (Chibu et al., 2002). However, results were similarly reported by Boonlertnirun et al. (2005) in rice and Rehim et al. (2009) in maize and bean. Among pulses, the use of glyphosate concentration above $3000 \mu\text{g ml}^{-1}$ not only affects plant growth and yield but is also responsible for genotoxicity (Sharma et al., 2019a,b; Liu et al., 2020).

4.4 Physiological responses of plants to pesticide toxicity

Excessive usage of pesticides induces toxicity and symptoms can visible in the form of necrosis, chlorosis, stunting, burns, and twisting of leaves, etc. (Sharma et al., 2018a; Donald, 2004). Toxicity depends upon various factors including rate of application, spraying technique, use of pesticides, climate conditions, floral organization, humidity, soil texture pH, moisture, temperature, and microbial interaction (Sharma et al., 2015a,b,c, 2016a). Pesticide application adversely affects the plant growth and development reported by Shahzad et al. (2018).

According to Sharma et al. (2018b), the application of pesticides induces oxidative injuries induces by ROS production. Oxidative stress promotes the degradation of proteins and chlorophyll pigments, which ultimately decrease the photosynthetic efficiency (Xia et al., 2006; Sharma et al., 2015a,b,c). To manage the damage of oxidative stress activates an antioxidative defense mechanism that involves a nonenzymatic and enzymatic antioxidants system (Xia et al., 2009; Sharma et al., 2016). According to Sharma et al., 2017, ROS scavenging elicited by activation of the antioxidative defense system aids with reduction of the oxidative stress in plants induced pesticide toxicity.

Pesticide application results in oxidative stress to plants by the production of ROS which then led to decrease photosynthetic efficiency and retarded growth in the plant. Some examples of physiological changes in plants after pesticide treatments are well tabulated in Tables 2–4.

TABLE 2 Impact of pesticides application on reactive oxygen species in different plants species.

S. no.	Pesticides used	Crop	Application mode	Pesticide concentration	Treatment time	Plant parts	Effect of pesticide on reactive oxygen species		References
							Parameter	Effects	
1.	Imidacloprid	Rice	Soil	0.015%	12 days	Seedling	O ₂ ⁻ H ₂ O ₂ MDA	Increase Increase Increase	Sharma et al. (2015a,b,c)
2.	Chlorpyrifos	Rice	Soil	0.04%	12 days	Seedling	O ₂ ⁻ H ₂ O ₂ MDA	Increase Increase Increase	Sharma et al. (2015a,b,c)
3.	Deltamethrin	Soybean	Spray	0.20%	10 days	Leaves	MDA	Increase	Bashir et al. (2007)
4.	Chlorpyrifos	Mung Bean	Spray	15 mM	10 days	Leaves	TBARS	Increase	Parween et al. (2012)
5.	Dimethoate	Mung Bean	Culture solution	150 ppm	4 days	Leaves	O ₂ ⁻ H ₂ O ₂ MDA OH ⁻	Increase Increase Increase Increase	Singh et al. (2014)
6.	Glyphosate	Soybean	Foliar spray	~1%	1 day	Leaves	MDA	Increase	Li et al. (2019)
7.	Clodinafop-Propargyl	Wheat	Foliar spray	800 µg/L	7 days	Leaves	O ₂ ⁻	Increased	Aliverdi et al. (2009)
8.	Paraquat	Opium poppy	Foliar spray	~50%	1 day	Leaves	MDA	Increase	Yuan et al. (2017)
9.	Atrazine	Rice	Hoagland solution	0.4 mg/L	6 days	Leaves	H ₂ O ₂	Increase	Zhang et al. (2014)
10.	Chlorotoluron	Wheat	Soil	25 mg/kg	10 days	Leaves	H ₂ O ₂	Increase	Song et al. (2010)
11.	Granstar	Oat	Spray	300 µg/L	3 days	Leaves	MDA	Increase	Gar'kova et al. (2011)
12.	Isoproturon	Pea	Soil	10 mM	15 days	Leaves	MDA	Increase	Singh et al. (2016)
13.	Dichlorobenzene	Maize	Hoagland solution	80 mg/L	7 days	Roots	H ₂ O ₂	Increase	Zhang et al. (2018)
14.	Trichlorobenzene	Maize	Hoagland solution	80 mg/L	7 days	Leaves	H ₂ O ₂	Increase	San Miguel et al. (2012)

TABLE 3 Impact of different pesticides on the activity of enzymatic antioxidants among different plants.

S. no.	Pesticides used	Plant studied	Application mode	Concentration used	Time after treatment	Part of plant analyzed	Effect of pesticide on the activity of enzymatic antioxidants		References
							Parameter	Effects	
1.	Imidacloprid	Mustard	Soil	300 mg/kg	65 days	Leaves	GR GST POD	Increase Increase Increase	Sharma et al. (2016b)
					80 days	Green Pods	APOX GPOX GR GST POD	Increase Increase Increase Increase Increase	Sharma et al. (2017c)
2.	DDT	Soybean	Soil	63.5 ng/g	60 days	Root leaves	GST GST	Increase Decrease	Mitton et al. (2016)
3.	Chlorpyrifos	Mung Bean	Foliar spray	15 mM	10 days	Leaves	APOX GR SOD CAT	Increase Increase Increase Decrease	Parween et al. (2012)
4.	Deltamethrin	Soybean	Foliar spray	0.20%	10 days	Leaves	APOX GR SOD CAT	Increase Increase Increase Decrease	Bashir et al. (2007)
5.	Atrazine	Maize	Foliar spray	1.79 kg a.i. ha ⁻¹	12 days	Shoot	SOD APOX CAT POD	Increase Decrease Decrease Decrease	Hassan and Alla (2005)
6.	Glyphosate	Soybean	Spray	~1%	1 day	Root	CAT POX	Increase Increase	Pall (2019)
7.	Fusilade	Peanut	Spray	60 ppm	14 days	Leaves	POD CAT	Increase Decrease	Osman et al. (2016)

8.	Fluroxypyr	Rice	Culture solution	0.8mg/L	6 days	Leaves	POD SOD	Increase Decrease	Zhang et al. (2014)
9.	Isoproturon	Wheat	Soil	4mg/kg	10 days	Roots	GR POD SOD	Increase Increase Increase	Liang et al. (2012)
10.	Dichlorobenzene	Maize	Hoagland solution	80mg/L	7 days	Leaves	POD GR	Decrease Decrease	San Miguel et al. (2012)
11.	Monochlorobenzene	Maize	Hoagland solution	80mg/L	7 days	Roots	POD GR	Increase Decrease	San Miguel et al. (2012)
12.	Chlorotoluron	Wheat	Soil	25mg/ Kg	10 days	Roots	SOD CAT	Increase Decrease	Song et al. (2010)
13.	Granstar	Wheat	Leaf disk	300µg/L	3h	Leaves	CAT APX	Increase Increase	Sharma et al. (2019a,b)

TABLE 4 Effect of pesticides on nonenzymatic antioxidants in plants.

S. no.	Pesticide name	Mode of application	Pesticide concentration	Plant name	Treatment time	Plant part analyzed	Effect of pesticide on oxidative stress markers		References
							Parameter	Effect	
1.	Chlorantraniliprole	Seed	0.5 ppm	Maize	7 days	Leaves	Proline	Increase	Kilic et al. (2015)
2.	Imidacloprid	Sand	0.02%	Rice	12 days	Seedlings	Proline	Increase	Sharma et al. (2013)
3.	Pendimethalin	Soil	8.5 mg/L	Fennel	84 days	Leaves	Total phenolics	Increase	El-Awadi and Hassan (2011)
4.	Chlorpyrifos	Spray	15 mM	Mungbean	10 days	Leaves	Proline	Increase	Parween et al. (2018)
5.	Deltamethrin	Spray	0.20%	Soybean	10 days	Leaves	Ascorbate GSH	Increase Increase	Sharma et al. (2019a,b)
6.	Atrazine	Soil	1.79 kg/ha	Maize	8 days	Shoot	Ascorbate GSH	Increase Decrease	Akbulut and Yigit (2010)
7.	Diuron	Soil	2 ppm	Soybean	7 days	Leaves	Proline	Increase	Fayez (2000)
8.	Bentazon	Spray	1.6 g/L	Peanut	14 days	Leaves	Proline	Increase	Fayez et al. (2013)
9.	Imazapic	Spray	0.12 mM	Tobacco	9 days	Leaves	GSH	Increase	Kaya and Doganlar (2016)
10.	Fomesafen	Spray	1000 g/ha	Soybean	2 days	Leaves	GSH	Increase	Matzenbacher et al. (2014)
11.	Oxyfluorfen	Spray	2500 g/ha	Soybean	2 days	Leaves	GSH	Increase	Benekos et al. (2010)

Pesticides affect crop physiology by different disruptions reproductive organs development, alteration in nutrients metabolism, growth reduction. The physiological disruptions solely rely upon the pesticide specificity and their mode of application (Petit et al., 2012). It abrupt photophosphorylation, photosynthesis process, electron flow, lipids solubilization and conformational changes (Murthy, 1983; Krugh and Miles, 1996) and/or mechanical effects (Ferree, 1979). Adverse effects of various pesticides applications on photosynthesis have been reported in lettuce (Toscano et al., 1982; Haile et al., 2000) cotton (Youngman et al., 1990), alfalfa (Haile et al., 1999), citrus (Jones et al., 1983), strawberry (Laprè et al., 1982), maize (Godfrey and Holtzer, 1992), peach (Andersen et al., 1986), soybean (Abdel-Reheem et al., 1991), azalea (Kingleman et al., 2000) and pecan (Wood and Payne, 1986). On the contrary, no considerable effect of excessive insecticide exposure/toxicity on photosynthetic rate (Krugh and Miles, 1996; Haile et al., 1999; Lloyd and Krieg, 1987).

Acetamiprid, imidacloprid, abamectin, thiamethoxam, and abamectin + chlorantraniliprole were applied on Hazera 5656 F1 variety of (*L. esculentum* Mill.) tomato under greenhouse conditions in Köyceğiz region of Muğla. It was found that the use of higher doses of pesticides adversely affected the physiological and biochemical properties of tomato plants. MDA (malondialdehyde), proline, H_2O_2 , SOD, POD contents, and CAT activities increased with the increase in pesticide doses. Increasing pesticides dose decreased the Dry Matter %, total chlorophyll, and carotenoid contents. Plants on which ABAC-3 was sprayed contained 56% proline content when compared with the control plants. Samples treated with ABA-3 (Abamectin) showed a maximum increase in superoxide dismutase (SOD) activities and the least decrease was shown by samples treated with THM-1 (Trihalomethanes). In maximum cases, the highest doses of applied pesticides increased CAT and POD activities (Yildiztekin et al., 2019).

Giménez-Moolhuyzen et al. (2020) tested the effect of 22 pesticides (commonly used insecticides and acaricides) on the photosynthetic activity of capsicum leaves during the initial stage of adoption of Integrated Pest Management (IPM). In this IPM system, 9 out of 22 studied pesticides showed a reduction in the photosynthesis activities of capsicum leaves after a single application. Whereas impact on the photosynthesis rate and duration of the impact varied for different pesticides. Among the studied phytosanitary products two types of effects on photosynthesis were observed. The first one is a transient effect, immediately after the treatment, possibly caused by a short-term effect on the stomatal conductance. Phytosanitary products which have this type of effect include vegetable oil (Addit) and an organic soap (Biosoap) (Giménez-Moolhuyzen et al., 2020; Winnicki et al., 2021).

Parween et al. (2012) investigated the effect of chlorpyrifos insecticides on the antioxidative enzymatic system of *V. radiata* L. metabolism. The enhanced rate of proline content and lipid peroxidation whereas the level of glutathione and declined significantly during the different developmental phases. In *Brassica juncea*, activities of antioxidant enzymes increased with the application of Imidacloprid insecticide (Sharma et al., 2015a,b,c).

Raja et al. (2013) investigated to find out the effect of endosulfan at the various amount of 0, 50, 100, 200, 400, and 600 ppm on photosynthetic pigments, ascorbic acid, dry weight, fresh weight, stress proteins, enzymes (nitrate reductase) and nitrate uptake on water fern, *Azolla microphylla*. The dose-dependent effect of endosulfan was noticed. Endosulfan affected the cellular activities which led to the increase and gradually decrease in Vitamin-C content at lower concentrations and higher concentrations, respectively. A decline in protein

concentration was also noticed whereas metabolic activities and nitrate uptake enhanced up to a particular level and got declined at higher concentrations. The total protein content of PGPRs (plant growth-promoting bacteria) was maximally decreased by 80% at a three-fold increase in kitazin, indicating its inhibitory effect on the bacterial protein synthesis mechanism (Shahid et al., 2020).

5. Other measures for the prevention of pesticides phytotoxicity

Chemical control is not the only option to control pests. There are other measures like cultural, mechanical, physical, and biological control, etc. also for controlling the pests. Do not rely only on the chemical control of pests. Plant protection measures other than chemical control should be adopted. Cultural control measures are the cornerstones of pest management in a protected environment; chemical controls cannot overcome poor or sloppy management. Numerous deleterious effects of pesticide toxicity can be mitigated by using alternative but efficient pest management approaches. However, in controlled growth chambers for horticulture, pest control depends upon parasitoids (arthropod) can causes better options over conventional synthetic agrochemicals (Van Lenteren, 2007). Furthermore, pesticides nanocarriers systems also play an imperative role in protecting our crops from fungal diseases. In this context, Maluin et al. (2020) conducted a study in which chitosan was used as nanocarriers for dazomet or hexaconazole-like fungicide in preparing a new agronanofungicides generation with an improved antifungal potent agent that exhibited no phytotoxic effect on plant and environment.

Results revealed that chitosan-fungicides-based nanoparticles significantly reduced the phytotoxic effect of fungicides when compared with their counterparts, pure fungicides in oil palm seedlings (Sangwan et al., 2023). In addition, the use of chitosan-fungicides-based nanoparticles is also a reliable approach to reduce the fungicide's phytotoxicity. These chitosan-agronanofungicides were observed to alleviate the photosynthetic rate in the seedling of oil palm. Chitosan nanoparticles are also used as an antifungal agent on *Ganoderma boninense*, which makes it a promising candidate in the regulation and management of stem rot disease in the basal region of oil palms (Maluin et al., 2019).

6. Conclusion

The ever-increasing population of the world is exerting pressure on land as well as agriculture which is threatening food security in the coming future. Insect pests, diseases, and weeds are responsible for lowering agriculture production. Therefore, to manage them and to fulfill the demand for food human beings are using chemical pesticides on their crops. But several studies have shown that pesticide application leads to the development of phytotoxicity in plants and it impairs the growth, production, and active constituents of plants. Seed germination, photosynthesis, and oxidants levels also get affected due to phytotoxicity. Pesticides will remain an imperative tool for the management of insect pests and play an important role in optimizing yields from the crops. Thus, the development of new pesticide molecules with lesser phytotoxic effect on crops is the need of the hour. In addition to these growers have to

use pesticides safely at the recommended dose and on recommended crops after following all the given guidelines.

Furthermore, users should not only rely on chemical control of insect pests, diseases, and weeds. Farmers should opt for other control measures like cultural, mechanical, physical, and biological control also. By doing this not only the plant yield will be improved but it will also benefit the environment and human health. Furthermore, future investigation is required in the regard to generate more reliable data and prepare a comprehensive model.

References

- Abdel-Reheem, S., Belal, M.H., Gupta, G., 1991. Photosynthesis inhibition of soybean leaves by insecticides. *Environ. Pollut.* 74, 245–250.
- Akbulut, G.B., Yigit, E., 2010. The changes in some biochemical parameters in *Zea mays* cv. “Martha F1” treated with atrazine. *Ecotoxicol. Environ. Saf.* 73, 1429–1432.
- Al-Ani, M.A.M., Hmoshi, R.M., Kanaan, I.A., Thanoon, A.A., 2019. Effect of pesticides on soil microorganisms. *J. Phys. Conf. Ser.* 1294, 072007.
- Aliverdi, A., Mohassel, M.H.R., Zand, E., Mahallati, M.N., 2009. Increased foliar activity of clodinafop-propargyl and/or tribenuron-methyl by surfactants and their synergistic action on wild oat (*Avena ludoviciana*) and wild mustard (*Sinapis arvensis*). *Weed Biol. Manag.* 9, 292–299.
- Aluthmuhandiram, J.V.S., Chethana, K.W.T., Zhang, W., Peng, J., Zhao, E., Li, X.H., Saichana, N., Yan, J., 2021. Impact of temperature variation on the phytotoxic secondary metabolite production by *Lasiodiplodia theobromae*. *J. Phytopathol.* <https://doi.org/10.1111/jph.13043>.
- Amin, A.W., 2002. Cytotoxicity testing of sewage water treatment using *Allium cepa* chromosome aberration assay. *Pak. J. Biol. Sci.* 5, 184–188.
- Andersen, P.C., Mizell, R.F., French, W.J., Aldrich, J.H., 1986. Effect of multiple applications of pesticides on leaf gas exchange of peach. *Hortic. Sci.* 21, 508–510.
- Bačmaga, M., Wyszowska, J., Kucharski, J., 2016. The effect of the Falcon 460 EC fungicide on soil microbial communities, enzyme activities and plant growth. *Ecotoxicology* 25, 1575–1587.
- Bačmaga, M., Wyszowska, J., Kucharski, J., 2018. The influence of chlorothalonil on the activity of soil microorganisms and enzymes. *Ecotoxicology* 27, 1188–1202.
- Balikai, R.A., Mallapur, C.P., 2017. Bio-efficacy of flubendiamide 480 SC (Fame 480 SC) against fruit borer in Gherkin. *Int. J. Horticult.* 7, 250–261.
- Bashir, F., Mahmooduzzafar, Siddiqi, T.O., Iqbal, M., 2007. The antioxidative response system in *Glycine max* (L.) Merr. exposed to deltamethrin, a synthetic pyrethroid insecticide. *Environ. Pollut.* 147, 94–100.
- Benekos, K., Kissoudis, C., Nianiou-Obeidat, I., Labrou, N., Madesis, P., Kalamaki, M., Tsafaris, A., 2010. Overexpression of a specific soybean GmGSTU4 isoenzyme improves diphenyl ether and chloroacetanilide herbicide tolerance of transgenic tobacco plants. *J. Biotechnol.* 150, 195–201.
- Biondi, A., Desneux, N., Siscaro, G., Zappalà, L., 2012. Using organic-certified rather than synthetic pesticides may not be safer for biological control agents: selectivity and side effects of 14 pesticides on the predator *Orius laevis*. *Chemosphere* 87, 803–812.
- Boonlertnirun, S., Boonlertnirun, K., Sooksathan, I., 2005. Proceedings of 43rd Kasetsart University Annual Conference, Thailand, 1–4 February. pp. 37–43.
- Casida, J.E., Quistad, G.B., 1998. Golden age of insecticide research: past, present, or future? *Annu. Rev. Entomol.* 43, 1–16.
- Chibu, H., Shibayama, H., Arimas, S., 2002. Effects of chitisan application on the shoot growth of rice and soybean. *Japan. J. Crop Sci.* 71, 206–211.
- Dhanamanjuri, W., Thoudam, R., Dutta, B.K., 2013. Effect of some pesticides (fungicides) on the germination and growth of seeds/seedlings of some crop plants (i.e. *Cicer arietinum* and *Zea mays*). *Middle-East J. Sci. Res.* 17, 627–632.
- Donald, P.F., 2004. Biodiversity impacts of some agricultural commodity production systems. *Conserv. Biol.* 18, 17–37.
- Dubey, K.K., Fulekar, M.H., 2011. Effect of pesticides on the seed germination of *Cenchrus setigerus* and *Pennisetum pedicellatum* as monocropping and co-cropping system: implications for rhizospheric bioremediation. *Roman. Biotechnol. Lett.* 16, 5909–5918.

- Ebel, R.C., Wallace, B., Elkins, C., 2000. Phytotoxicity of the systemic insecticide imidacloprid on tomato and cucumber in the greenhouse. *HortTechnology* 10, 144–147.
- El-Awadi, M.E., Hassan, E.A., 2011. Improving growth and productivity of fennel plant exposed to pendimethalin herbicide: stress–recovery treatments. *Nat. Sci.* 9, 97–108.
- Fatma, F., Verma, S., Kamal, A., Srivastava, A., 2017. Phytotoxicity of pesticides mancozeb and chlorpyrifos: correlation with the antioxidative defence system in *Allium cepa*. *Physiol. Mol. Biol. Plants* 24, 115–123.
- Fatma, F., Verma, S., Kamal, A., Srivastava, A., 2018. Phytotoxicity of pesticides mancozeb and chlorpyrifos: correlation with the antioxidative defence system in *Allium cepa*. *Physiol. Mol. Biol. Plants* 24, 115–123.
- Fayez, K.A., 2000. Action of photosynthetic diuron herbicide on cell organelles and biochemical constituents of the leaves of two soybean cultivars. *Pestic. Biochem. Physiol.* 66, 105–115.
- Fayez, K., Radwan, D., Khalaf, A., Abdelrahman, A.M., 2013. Herbicides and salicylic acid applications caused alterations in total amino acids and proline contents of peanut cultivars. *J. Environ. Stud.* 6, 55–61.
- Fernandes, J.P.T., Nascente, A.S., Filippi, M.C.C., Silva, M.A., 2021. Upland rice seedling performance promoted by multifunctional microorganisms. *Semina: Ciênc. Agrár.* 1, 429–438.
- Ferree, D.C., 1979. Influence of pesticides on photosynthesis of crop plants. In: Marcelle, R., Clijsters, H., van Poucke, M. (Eds.), *Photosynthesis and Plant Development*. Junk, The Hague, The Netherlands, pp. 331–341.
- Gar'kova, A.N., Rusaeva, M.M., Nushtaeva, O.V., Aroslinkina, Y.N., Lukatkin, A.S., 2011. Treatment with the herbicide granstar induces oxidative stress in cereal leaves. *Russ. J. Plant Physiol.* 58, 1074–1081.
- Giménez-Moolhuyzen, M., Blom, J., Lorenzo-Mínguez, P., Cabello, T., Crisol-Martínez, E., 2020. Photosynthesis inhibiting effects of pesticides on sweet pepper leaves. *Insects* 11, 69.
- Godfrey, L.D., Holtzer, T.O., 1992. Effects of soil-incorporated insecticides and foliar applied chemicals on corn gas-exchange parameters. *Crop Prot.* 11, 427–432.
- Goh, W.L., Yiu, P.H., Wong, S.C., Rajan, A., 2011. Safe use of chlorpyrifos for insect pest management in leaf mustard (*Brassica juncea* L. Coss.). *J. Food Agric. Environ.* 9, 1064–1066.
- Goswami, M.R., Banerjee, P., Swarnakar, S., Mukhopadhyay, A., 2013. Carbaryl mediated biochemical alterations in eggplant (*Solanum melongena* L.). *Int. J. Environ. Sci. Technol.* 3, 51.
- Habtamu, A., Shelema, M., Kedar, R., Ebsa, S., 2013. Seed germination and seedling growth of Haricot bean (*Phaseolus vulgaris*) cultivars as influenced by copper sulphate. *World Sci. Res. J.* 10, 312–317.
- Haile, F.J., Peterson, R.K.D., Higley, L.G., 1999. Gas-exchange responses of alfalfa and soybean treated with insecticides. *J. Econ. Entomol.* 92, 954–959.
- Haile, F.J., Kerns, D.L., Richardson, J.M., Higley, L.G., 2000. Impact of insecticides and surfactant on lettuce physiology and yield. *J. Econ. Entomol.* 93, 788–794.
- Han, Y., Mo, R., Yuan, X., Zhong, D., Tang, F., Ye, C., Liu, Y., 2017. Pesticide residues in nut-planted soils of China and their relationship between nut/soil. *Chemosphere* 180, 42–47.
- Hasan, W., Chhibber, R.C., Singh, C.P., 2016. Effect of indoxacarb against tomato fruit borer (*Helicoverpa armigera* Hub.) and phytotoxicity to tomato plants. *Adv. Plants Agric. Res.* 3, 51–54.
- Hassaan, M., El Nemr, A., 2020. Pesticides pollution: classifications, human health impact, extraction and treatment techniques. *Egypt. J. Aquat. Res.* 46. <https://doi.org/10.1016/j.ejar.2020.08.007>.
- Hassan, N.M., Alla, M.M.N., 2005. Oxidative stress in herbicide-treated broad bean and maize plants. *Acta Physiol. Plant.* 27, 429–438.
- Homayoonzadeh, M., Moeini, P., Talebi, K., Roessner, U., Hosseiniaveh, V., 2020. Antioxidant system status of cucumber plants under pesticides treatment. *Acta Physiol. Plant.* 42, 161. <https://doi.org/10.1007/s11738-020-03150-9>.
- Homayoonzadeh, M., Hosseiniaveh, V., Haghighi, S.R., Talebi, K., Roessner, U., Maali-Amiri, R., 2021. Evaluation of physiological and biochemical responses of pistachio plants (*Pistacia vera* L.) exposed to pesticides. *Ecotoxicology*. <https://doi.org/10.1007/s10646-021-02434-1>.
- Jan, S., Parween, T., Siddiqi, T.O., Mahmooduzzafar, 2012. Effect of gamma radiation on morphological, biochemical and physiological aspects of plants and plant products. *Environ. Rev.* 20, 7–39.
- Jiang, L., Yang, H., 2009. Prometryne-induced oxidative stress and impact on antioxidant enzymes in wheat. *Ecotoxicol. Environ. Saf.* 72, 1687–1693.
- Jianga, L., Maa, L., Suia, Y., Hna, S.Q., Wua, Z.Y., Fenga, Y.X., Yanga, H., 2010. Effect of manure compost on the herbicide prometryne bioavailability to wheat plants. *J. Hazard. Mater.* 184, 337–344.
- Jones, V.P., Youngman, R.R., Parrella, M.P., 1983. Effect of selected acaricides on photosynthetic rates of lemon and orange leaves in California. *J. Econ. Entomol.* 76, 1179–1180.

- Kaur, R., Kaur, G.M., Raghav, S., 2019. Pesticides classification and its impact on environment. *Int. J. Curr. Microbiol. App. Sci.* 8, 1889–1897.
- Kaya, A., Doganlar, Z.B., 2016. Exogenous jasmonic acid induces stress tolerance in tobacco (*Nicotiana tabacum*) exposed to imazapic. *Ecotoxicol. Environ. Saf.* 124, 470–479.
- Kilic, S., Duran, R.E., Coskun, Y., 2015. Morphological and physiological responses of maize (*Zea mays* L.) seeds grown under increasing concentrations of chlorantraniliprole insecticide. *Pol. J. Environ. Stud.* 24, 1069–1075.
- Kingleman, W.E., Buntin, G.D., van Iersel, M.W., Braman, S.K., 2000. Whole plant gas exchange, not individual leaf-measurements, accurately assesses azalea response to insecticides. *Crop Prot.* 19, 407–415.
- Koornneef, M., Bentsink, L., Hilhorst, H., 2002. Seed dormancy and germination. *Curr. Opin. Plant Biol.* 5, 33.
- Krugh, B.W., Miles, D., 1996. Monitoring the effects of five “nonherbicidal pesticide chemicals on terrestrial plants using chlorophyll fluorescence”. *Environ. Toxicol. Chem.* 15, 495–500.
- Kumar, A., Sharma, A., Sharma, R., Srivastava, P., Choudhary, A., 2021a. Exploration of wheat wild relative diversity from Lahaul valley: a cold arid desert of Indian Himalayas. *Cereal Res. Commun.* <https://doi.org/10.1007/s42976-021-00166-w>.
- Kumar, A., Sharma, A., Sharma, R., Choudhary, A., Srivastava, P., Kaur, H., Padhy, A.K., 2021b. Morpho-physiological evaluation of *Elymus semicostatus* (Nees ex Steud.) Melderis as potential donor for drought tolerance in wheat (*Triticum aestivum* L.). *Genet. Resour. Crop. Evol.* <https://doi.org/10.1007/s10722-021-01241-1>.
- Kumar, A., Choudhary, A., Kaur, N., Kaur, H., 2021c. Wake up: it's time to bloom. *Russ. J. Plant Physiol.* 68, 579–595.
- Kumar, A., Choudhary, A., Kaur, H., Mehta, S., Husen, A., 2021d. Metal-based nanoparticles, sensors, and their multifaceted application in food packaging. *J. Nanobiotechnol.* 19, 256.
- Lal, S.K., Kumar, S., Sheri, V., Mehta, S., Varakumar, P., Ram, B., Borphukan, B., James, D., Fartyal, D., Reddy, M.K., 2018. Seed priming: an emerging technology to impart abiotic stress tolerance in crop plants. In: *Advances in Seed Priming*. Springer, Singapore, pp. 41–50.
- Laprè, L.F., Sances, F.V., Toscano, N.C., Oatman, E.R., Voth, V., Johnson, M.W., 1982. The effects of acaricides on the physiology, growth, and yield of strawberries. *J. Econ. Entomol.* 75, 616–619.
- Li, W.Y., Lu, P., Xie, H., Li, G.Q., Wang, J.X., Guo, D.Y., Liang, X.Y., 2019. Effects of glyphosate on soybean metabolism in strains bred for glyphosate-resistance. *Physiol. Mol. Biol. Plants* 25, 523–532.
- Liang, L., Lu, Y.L., Yang, H., 2012. Toxicology of isoproturon to the food crop wheat as affected by salicylic acid. *Environ. Sci. Pollut. Res.* 19, 2044–2054.
- Liu, W., Li, Y., Feng, Y., 2020. The effectiveness of nanobiochar for reducing phytotoxicity and improving soil remediation in cadmium-contaminated soil. *Sci. Rep.* 10, 858.
- Lloyd, R.W., Krieg, D.R., 1987. Cotton development and yield as affected by insecticides. *J. Econ. Entomol.* 80, 854–858.
- Lushchak, V.I., Matviishyn, T.M., Husak, V.V., Storey, J.M., Storey, K.B., 2018. Pesticide toxicity: a mechanistic approach. *EXCLI J.* 17, 1101–1136.
- Maluin, F., Hussein, M., Yusof, N., Fakurazi, S., Abu, S.I., Zainol, H., Nur, H., Daim, L., 2019. Enhanced fungicidal efficacy on *Ganoderma boninense* by simultaneous co-delivery of hexaconazole and dazomet from their chitosan nanoparticles. *RSC Adv.* 9, 27083–27095.
- Maluin, F.N., Hussein, M.Z., Yusof, N.A., Fakurazi, S., Idris, A.S., Hilmi, N.H.Z., 2020. Phytotoxicity of chitosan-based agronofungicides in the vegetative growth of oil palm seedling. *PLoS One* 15, e0231315.
- Matzenbacher, F.D.O., Vidal, R.A., Merotto Jr., A., Trezzi, M.M., 2014. Environmental and physiological factors that affect the efficacy of herbicides that inhibit the enzyme protoporphyrinogen oxidase: a literature review. *Planta Daninha* 32, 457–463.
- Mehta, S., Gogna, M., Singh, B., Patra, A., Singh, I.K., Singh, A., 2020. Silicon: a plant nutritional “non-entity” for mitigating abiotic stresses. In: *Plant Stress Biology*. Springer, Singapore, pp. 17–49.
- Mehta, S., Kumar, A., Achary, V.M.M., Ganesan, P., Patel, A., Singh, A., Rath, N., Das, T.K., Lal, S.K., Reddy, M.K., 2022. Antifungal and defense elicitor activity of Potassium phosphite against fungal blast disease on ptxD-OE transgenic indica rice and its acceptor parent. *Pestic. Biochem. Physiol.* 182, 105026.
- Mehta, S., Kumar, A., Achary, V.M.M., Ganesan, P., Rath, N., Singh, A., Sahu, K.P., Lal, S.K., Das, T.K., Reddy, M.K., 2021. Antifungal activity of glyphosate against fungal blast disease on glyphosate-tolerant OsmEPSPS transgenic rice. *Plant Sci.* 311, 111009.
- Mehta, S., Singh, B., Dhakate, P., Rahman, M., Islam, M.A., 2019. Rice, marker-assisted breeding, and disease resistance. In: *Disease Resistance in Crop Plants*. Springer, Cham, pp. 83–111.

- Mei, X., Lin, D.H., Xu, Y., Wu, Y.Y., Tu, Y.Y., 2009. Effects of phenanthrene on chemical composition and enzyme activity in fresh tea leaves. *Food Chem.* 115, 569–573.
- Mishra, V., Srivastava, G., Prasad, S.M., 2009. Antioxidant response of bitter gourd (*Momordica charantia* L.) seedlings to interactive effect of dimethoate and UVB irradiation. *Sci. Hortic.* 120, 373–378.
- Mitton, F.M., Ferreira, J.L.R., Gonzalez, M., Miglioranza, K.S.B., Monserrat, J.M., 2016. Antioxidant responses in soybean and alfalfa plants grown in DDTs contaminated soils: useful variables for selecting plants for soil phytoremediation? *Pestic. Biochem. Physiol.* 130, 17–21.
- Monika, Kidwai, M.K., 2017. Effect of mancozeb on mustard (*Brassica juncea* L.): an *in-vitro* study. *Trop. Plant Res.* 4, 55–61.
- Moore, M.T., Kroger, R., 2010. Effect of three insecticides and two herbicides on rice (*Oryza sativa*) seedling germination and growth. *Arch. Environ. Contam. Toxicol.* 59, 574–581.
- Murthy, C.S.H.N., 1983. Effect of pesticides on photosynthesis. *Residue Rev.* 86, 107–129.
- Nannipieri, P., Giagnoni, L., Renella, G., Puglisi, E., Ceccanti, B., Masciandaro, G., Fornasier, F., Moscatelli, M.C., Marinari, S., 2013. Soil enzymology: classical and molecular approaches. *Biol. Fertil. Soils* 48, 743–762.
- Nansen, C., Purington, R., Murdock, M., 2021. Using advanced optical sensing to quantify phytotoxicity in ornamental plants. *HortTechnology* 31, 532–534.
- Nehru, S.D., Rangaiah, S., Ramarao, G., Shekar, G.C., 1999. Effect of some herbicide on seed germination and seedling vigour in mungbean. *Crop. Res.* 17, 425–426.
- Nguyen, H., Nansen, C., 2020. Hyperspectral remote sensing to detect leafminer-induced stress in bok choy and spinach according to fertilizer regime and timing. *Pest Manag. Sci.* 76, 2208–2216.
- Nielson, M.H., Rank, J., 1994. Screening of toxicity and genotoxicity in waste water by the use of *Allium* test. *Hereditas* 121, 249–254.
- Osman, M.E.H., Abo-Shady, A.M., El-Nagar, M.M., 2016. Cyanobacterial *Arthrospira* (*Spirulina platensis*) as safener against harmful effects of fusilade herbicide on faba bean plant. *Rendiconti Lincei* 27, 455–462.
- Pall, A., 2019. Mitigation of Glyphosate-Induced Plant Stress in Soybean Using Salicylic Acid. McGill University (Canada).
- Pandey, K.B., Rizvi, S.I., 2010. Markers of oxidative stress in erythrocytes and plasma during aging in humans. *Oxidative Med. Cell. Longev.* 3, 2–12.
- Parween, T., 2012. Effect of agrochemical pollution on growth, structure and some physiochemical aspects of *Vigna radiata* L. Ph.D thesis, (Awarded) Department of Biosciences, Jamia Millia Islamia, New Delhi, India.
- Parween, T., Jan, S., Mahmooduzzafar, Fatma, T., 2011a. Alteration in nitrogen metabolism and plant growth during different developmental stages of green gram (*Vigna radiata* L.) in response to chlorpyrifos. *Acta Physiol. Plant.* 33, 2321–2328.
- Parween, T., Jan, S., Mahmooduzzafar, Fatma, T., 2011b. Assessing the impact of chlorpyrifos on growth, photosynthetic pigments and yield in *Vigna radiata* L. at different phenological stages. *Afr. J. Agric. Res.* 6, 4432–4440.
- Parween, T., Jan, S., Fatma, M.T., 2012. Evaluation of oxidative stress in *Vigna radiata* L. in response to chlorpyrifos. *Int. J. Environ. Sci. Technol.* 9, 605–612.
- Parween, T., Jan, S., Fatma, T., 2018. Variation in elemental composition as influenced by chlorpyrifos application in mung bean (*Vigna radiata* L.). *Saudi J. Biol. Sci.* 25, 1439–1445.
- Petit, A.N., Fontaine, F., Vatsa, P., Clément, C., Vaillant-Gaveau, N., 2012. Fungicide impacts on photosynthesis in crop plants. *Photosynth. Res.* 111, 315–326.
- Rahman, M., Sultana, S., Nath, D., Kalita, S., Chakravarty, D., Mehta, S., Wani, S.H., Islam, M.A., 2019. Molecular breeding approaches for disease resistance in sugarcane. In: *Disease Resistance in Crop Plants*. Springer, Cham, pp. 131–155.
- Raja, W., Rathau, P., John, S.A., Ramteke, P., 2013. Photosynthetic, biochemical and enzymatic investigation of *Azolla microphylla* in response to an insecticide-Hexachlorohexahydro-Methano-Benzodioxathiepine-Oxide. *Our Nature* 10, 145–155.
- Rajashekhar, N., Prakasha, Murthy, T.C.S., 2012. Seed germination and physiological behavior of maize (cv. Nac-6002) seedlings under abiotic stress (Pendimethalin) condition. *Asian J. Crop Sci.* 4, 80–85.
- Reddy, B., Kumar, A., Mehta, S., Sahu, K.P., 2021. Methods of assessments of microbial diversity and their functional role in soil fertility and crop productivity. *Plant, Soil and Microbes in Tropical Ecosystems*. Springer, Singapore, pp. 293–314.
- Rehim, H.A.A., Hegazy, E.A., El-Barbary, A.M., 2009. Radiation modification of natural polysaccharides for application in agriculture. *Polymer* 50, 1952–1957.

- San Miguel, A., Faure, M., Ravanel, P., Raveton, M., 2012. Biological responses of maize (*Zea mays*) plants exposed to chlorobenzenes. Case study of monochloro-, 1, 4-dichloro- and 1, 2, 4-trichloro-benzenes. *Ecotoxicology* 21, 315–324.
- Sangwan, S., Sharma, P., Wati, L., Mehta, S., 2023. Effect of chitosan nanoparticles on growth and physiology of crop plants. In: *Engineered Nanomaterials for Sustainable Agricultural Production, Soil Improvement and Stress Management*. Academic Press, pp. 99–123.
- Shahid, M., Ahmed, B., Zaidi, A., Khan, M.S., 2018. Toxicity of fungicides to *Pisum sativum*: a study of oxidative damage, growth suppression, cellular death and morpho-anatomical changes. *RSC Adv.* 8, 38483–38498.
- Shahid, M., Khan, M.S., Zaidi, A., 2020. Fungicide toxicity to legumes and its microbial remediation: a current perspective. In: *Pesticides in Crop Production: Physiological and Biochemical Action*. Wiley, pp. 15–33.
- Shahzad, B., Tanveer, M., Che, Z., Rehman, A., Cheema, S.A., Sharma, A., Song, H., Rehman, S., Zhaorong, D., 2018. Role of 24-epibrassinolide (EBL) in mediating heavy metal and pesticide induced oxidative stress in plants: a review. *Ecotoxicol. Environ. Saf.* 147, 935–944.
- Shakir, S.K., Kanwal, M., Murad, W., Rehman, Z., Rehman, S., Daud, M.K., Azizullah, A., 2016. Effect of some commonly used pesticides on seed germination, biomass production and photosynthetic pigments in tomato (*Lycopersicon esculentum*). *Ecotoxicology*. <https://doi.org/10.1007/s10646-015-1591-9>.
- Sharma, I., Bhardwaj, R., Pati, P.K., 2013. Stress modulation response of 24-epibrassinolide against imidacloprid in an elite indica rice variety Pusa Basmati-1. *Pestic. Biochem. Physiol.* 105, 144–153.
- Sharma, I., Bhardwaj, R., Pati, P.K., 2015a. Exogenous application of 28-homobrassinolide modulates the dynamics of salt and pesticides induced stress responses in an elite rice variety Pusa Basmati-1. *J. Plant Growth Regul.* 34, 509–518.
- Sharma, A., Kumar, V., Kanwar, M.K., Thukral, A.K., Bhardwaj, R., 2017c. Ameliorating imidacloprid induced oxidative stress by 24-epibrassinolide in *Brassica juncea* L. *Russ. J. Plant Physiol.* 64, 509–517.
- Sharma, A., Kumar, V., Kohli, S.K., Thukral, A.K., Bhardwaj, R., 2015b. Phytochemicals in *Brassica juncea* L. seedlings under imidaclopridepibrassinolide treatment using GC-MS. *J. Chem. Pharm. Res.* 7, 708–711.
- Sharma, A., Kumar, V., Singh, R., Thukral, A.K., Bhardwaj, R., 2015c. 24-Epibrassinolide induces the synthesis of phytochemicals effected by imidacloprid pesticide stress in *Brassica juncea* L. *J. Pharmacogn. Phytochem.* 4, 60–64.
- Sharma, A., Kumar, V., Singh, R., Thukral, A.K., Bhardwaj, R., 2016a. Effect of seed pre-soaking with 24-epibrassinolide on growth and photosynthetic parameters of *Brassica juncea* L. in imidacloprid soil. *Ecotoxicol. Environ. Saf.* 133, 195–201.
- Sharma, A., Bhardwaj, R., Kumar, V., Thukral, A.K., 2016b. GC-MS studies reveal stimulated pesticide detoxification by brassinolide application in *Brassica juncea* L. plants. *Environ. Sci. Pollut. Res.* 23, 14518–14525.
- Sharma, A., Kumar, V., Kanwar, M.K., Thukral, A.K., Bhardwaj, R., 2017a. Ameliorating imidacloprid induced oxidative stress by 24-epibrassinolide in *Brassica juncea* L. *Russ. J. Plant Physiol.* 64, 509–517.
- Sharma, A., Kumar, V., Kumar, R., Shahzad, B., Thukral, A.K., Bhardwaj, R., 2018a. Brassinosteroid-mediated pesticide detoxification in plants: a mini-review. *Cogent Food Agric.* 4. <https://doi.org/10.1080/23311932.2018.1436212>.
- Sharma, A., Kumar, V., Yuan, H., Kanwar, M.K., Bhardwaj, R., Thukral, A.K., Zheng, B., 2018b. Jasmonic acid seed treatment stimulates insecticide detoxification in *Brassica juncea* L. *Front. Plant Sci.* 9, 1609. <https://doi.org/10.3389/fpls.2018.01609>.
- Sharma, A., Kumar, V., Shahzad, B., Tanveer, M., Sidhu, G.P.S., Handa, N., Kohli, S.K., Yadav, P., Bali, A.S., Parihar, R.D., 2019a. Worldwide pesticide usage and its impacts on ecosystem. *SN Appl. Sci.* 1, 1446.
- Sharma, A., Kumar, V., Thukral, A.K., Bhardwaj, R., 2019b. Responses of plants to pesticide toxicity: an overview. *Planta Daninha* 37.
- Singh, V.P., Kumar, J., Singh, S., Prasad, S.M., 2014. Dimethoate modifies enhanced UV-B effects on growth, photosynthesis and oxidative stress in mung bean (*Vigna radiata* L.) seedlings: implication of salicylic acid. *Pestic. Biochem. Physiol.* 116, 13–23.
- Singh, B., Mehta, S., Aggarwal, S.K., Tiwari, M., Bhuyan, S.I., Bhatia, S., Islam, M.A., 2019. Barley, disease resistance, and molecular breeding approaches. In: *Disease Resistance in Crop Plants*. Springer, Cham, pp. 261–299.
- Singh, H., Singh, N.B., Singh, A., Hussain, I., Yadav, V., 2016. Physiological and biochemical effects of salicylic acid on *Pisum sativum* exposed to isoproturon. *Arch. Agron. Soil Sci.* 62, 1425–1436.
- Snell Scientifics, LLC, 2017. Phytotoxicity of insecticides, fungicides and herbicides; symptoms, testing formulations & more. <https://www.snellsci.com/phytotoxicity-of-insecticides-fungicides-herbicides-symptoms-testing-formulations-more/>.

- Song, N.H., Zhang, S., Hong, M., Yang, H., 2010. Impact of dissolved organic matter on bioavailability of chlorotoluron to wheat. *Environ. Pollut.* 158, 906–912.
- Tevini, M., Teramura, A., 2008. UV-B effects on terrestrial plant. *Photochem. Photobiol.* 50, 479–487.
- Tomer, N., 2013. Determination of chlorinated pesticide in vegetables, cereals and pulses by gas chromatography in east national capital region, Delhi, India. *Res. J. Agric. Forest. Sci.* 1, 27–28.
- Toscano, N.C., Sances, M.W., Jonson, M.W., LaPré, L.F., 1982. Effect of various pesticides on lettuce physiology and yield. *J. Econ. Entomol.* 75, 738–741.
- Van Lenteren, J.C., 2007. Biological control for insect pests in greenhouses: an unexpected success. In: Charles Vincent, C., Goettel, M.S., Lazarovits, G. (Eds.), *Biological Control: A Global Perspective: Case Studies from Around the World*. CAB Int, Wallingford, UK, pp. 105–117.
- Wibawa, W., Mohamad, R.B., Puteh, A.B., Omar, D., Juraimi, A.S., Abdullah, S.A., 2009. Residual phytotoxicity effects of paraquat, glyphosate and glufosinate-ammonium herbicides in soils from field treated plots. *Int. J. Agric. Biol.* 11, 214–216.
- Winnicki, K., Łudzik, K., Żabka, A., 2021. Anti-algal activity of the 12-5-12 gemini surfactant results from its impact on the photosynthetic apparatus. *Sci. Rep.* 11, 2360.
- Wood, B.W., Payne, J.A., 1986. Net photosynthesis of orchard grown pecan leaves reduces by insecticide sprays. *Hortic. Sci.* 21, 112–113.
- Wu, G.L., Cui, E.J., Tao, E.L., Yang, E.H., 2010. Fluroxypyr triggers oxidative damage by producing superoxide and hydrogen peroxide in rice (*Oryza sativa*). *Ecotoxicology* 19, 24–132.
- Xia, X.J., Huang, Y.Y., Wang, L., Huang, L.F., Yu, W.L., Zhou, Y.H., Yu, J.Q., 2006. Pesticides induced depression of photosynthesis was alleviated by 24-epi-brassinolide pre-treatment in *Cucumis sativus* L. *Pestic. Biochem. Physiol.* 86, 42–48.
- Xia, X.J., Zhang, Y., Wu, J.X., Wang, J.T., Zhou, Y.H., Shi, K., 2009. Brassinosteroids promote metabolism of pesticides in cucumber. *J. Agric. Food Chem.* 57, 8406–8413.
- Yildiztekin, M., Ozler, M.A., Nadeem, S., Atilla Levent Tuna, A.L., 2019. Investigations on the effects of commonly used pesticides on tomato plant growth. *Fresenius Environ. Bull.* 28, 376–382.
- Youngman, R.R., Leigh, T.F., Kerby, T.A., Toscano, N.C., Jackson, C.E., 1990. Pesticides and cotton: effect on photosynthesis, growth, and fruiting. *J. Econ. Entomol.* 83, 1549–1557.
- Yuan, X.Y., Zhang, L.G., Huang, L., Yang, H.J., Zhong, Y.T., Ning, N., Guo, P.Y., 2017. Spraying brassinolide improves sigma broad tolerance in foxtail millet (*Setaria italica* L.) through modulation of antioxidant activity and photosynthetic capacity. *Sci. Rep.* 7, 1–9.
- Zhang, G.L., Chen, W.J., Qiu, L.M., Sun, G.R., Dai, Q.G., Zhang, H.C., 2009. Physiological response to 124-trichlorobenzene stress of different rice genotypes. *Acta Agron. Sin.* 35, 733–740.
- Zhang, J.J., Lu, Y.C., Zhang, J.J., Tan, L.R., Yang, H., 2014. Accumulation and toxicological response of atrazine in rice crops. *Ecotoxicol. Environ. Saf.* 102, 105–112.
- Zhang, S., Hu, X., Li, L., Huangfu, X., Xu, Y., Qin, Y., 2018. Activation of sodium percarbonate with ferrous ions for degradation of chlorobenzene in aqueous solution: mechanism, pathway and comparison with hydrogen peroxide. *Environ. Chem.* 14, 486–494.